

MTE-SLAM

Multi-Tier Feature Fusion for Efficient Neural Semantic SLAM

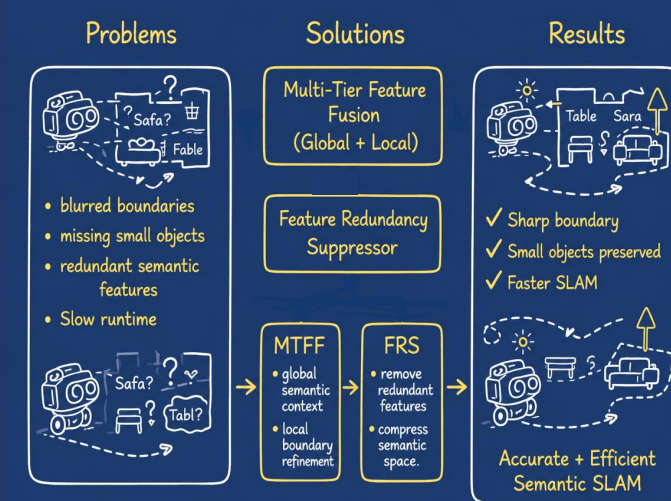
School of Artificial Intelligence and National Engineering Laboratory for Big Data System Computing Technology, Shenzhen University, Shenzhen, China.

Danqi Lu, Changxin Huang, Zhuangzhuang Chen, Zhiliang Lin, Dachong Li, Yanbin Chang, Jianqiang Li*



Motivation

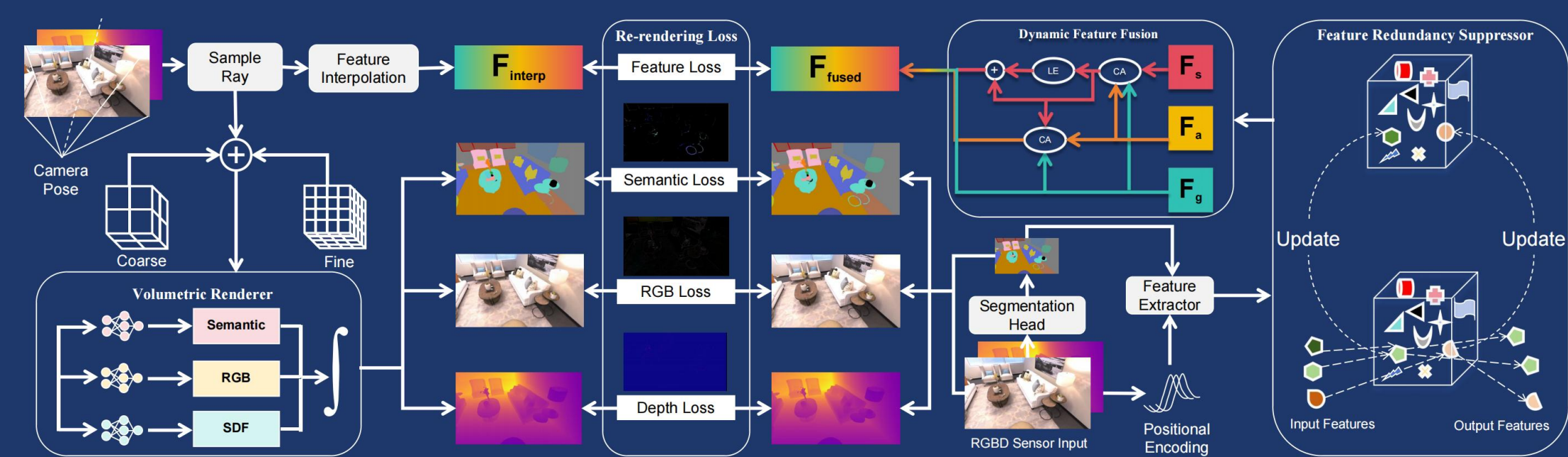
- Semantic SLAM jointly estimates ego-motion, builds global maps, and enables high-level scene understanding.
- Existing methods use simple semantic coloring or single-scale fusion, causing blurred boundaries and loss of fine structures.
- Redundant semantic features significantly reduce system efficiency.



Challenges

- Semantic underutilization.** Simple fusion strategies fail to fully exploit geometry, appearance, and semantic cues. They often miss global context and local boundary continuity, leading to coarse semantic maps and blurred object boundaries.
- Feature redundancy.** As frames accumulate, static semantic feature spaces store redundant features. This increases computation, reduces efficiency, and may degrade semantic map

Methods



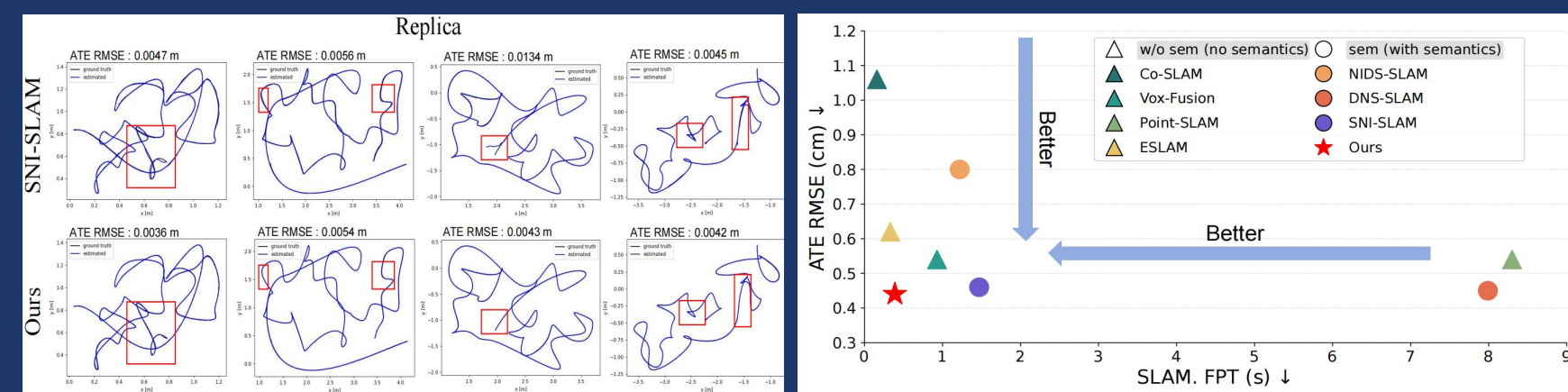
➤ MTFF: Multi-Tier Feature Fusion

- 1)Global Fusion: captures scene-level semantic context;
- 2)Local Refinement: sharpens boundaries and fine structures;
- 3)Adaptive Gating: balances geometry, semantics, and

➤ FRS: Feature Redundancy Suppressor

- 1)Prototype Anchors: provide stable semantic references;
- 2)Feature Compression: maps new features to nearest anchors;
- 3)Adaptive Update: refines anchors for compact representation.

Experiments

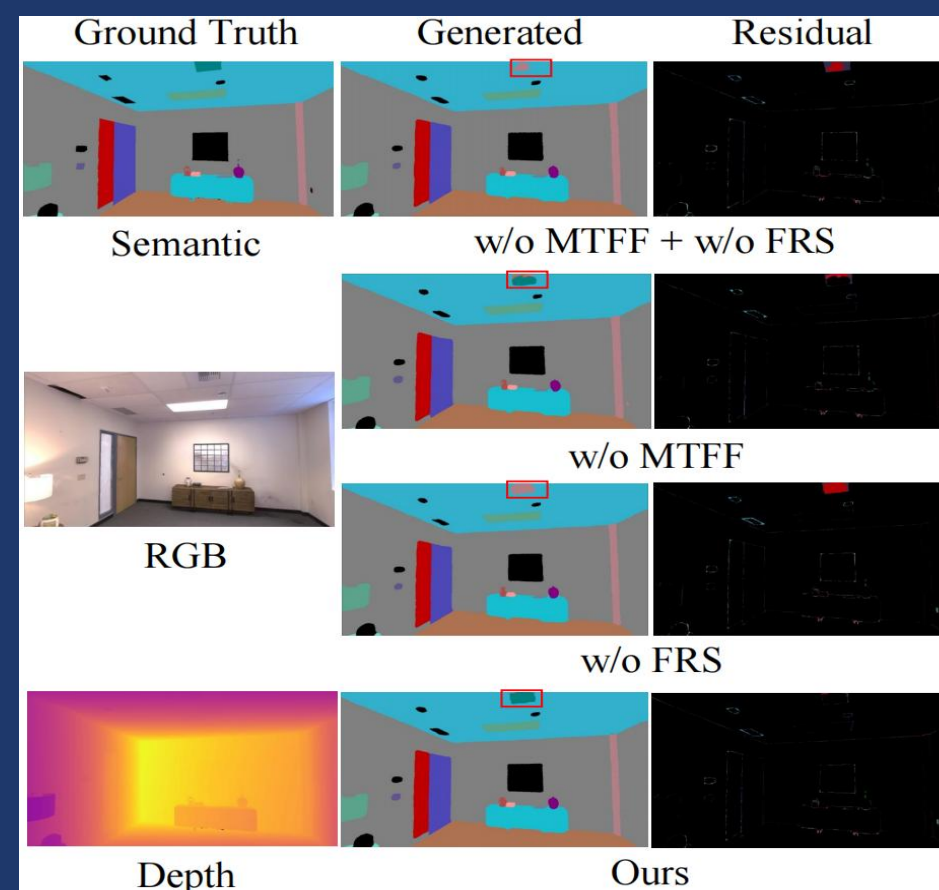


Qualitative Tracking Results

Accuracy-Efficiency Trade-off

Method	room0	room1	room2	office0	office1	office2	office3	office4
NIDS-SLAM	82.45	84.08	76.99	85.94	-	-	-	-
DNS-SLAM	88.32	84.90	81.20	84.66	-	-	-	-
SNI-SLAM	88.42	87.43	86.16	87.63	78.63	86.49	74.01	80.22
Ours	86.75	88.85	86.31	86.40	90.21	84.77	81.40	80.46

Semantic Segmentation mIoU (%) ↑



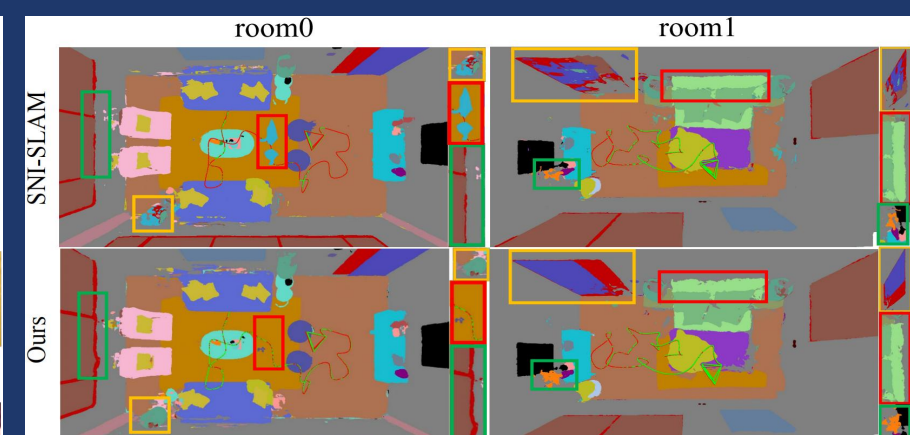
Qualitative Ablation Study



Qualitative RGB Reconstruction

Method/ATE RMSE (cm)	room0	room1	room2	office0
NICE-SLAM	1.86	2.37	2.26	1.50
Co-SLAM	0.72	0.85	1.02	0.69
QQ-SLAM	0.58	1.16	0.87	0.52
ESLAM	0.71	0.70	0.52	0.57
SNI-SLAM	0.50	0.55	0.45	0.33
Ours	0.55	0.44	0.43	0.36

Tracking Quality



Qualitative Semantic Reconstruction

Method	Depth L1 (cm) ↓	Acc. (cm) ↓	Comp. (cm) ↓	Comp. Ratio (%) ↑
NICE-SLAM	3.53	2.85	3.00	89.33
Co-SLAM	1.62	2.10	2.08	93.44
QQ-SLAM	1.42	2.43	1.93	94.38
ESLAM	1.18	0.97	1.05	98.60
SNI-SLAM	0.77	1.94	1.69	96.62
Ours	0.16	1.06	0.99	99.23

Reconstruction Quality